



**NORTH
SYDNEY
GIRLS HIGH
SCHOOL**

2022

**HSC
Assessment
Task 3**

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 60 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 6 – 8, show relevant mathematical reasoning and/or calculations

Total marks – 41

Section I – 5 marks (pages 2 – 4)

- Attempt Questions 1 – 5
- Allow about 7 minutes for this section

Section II – 36 marks (pages 5 – 7)

- Attempt Questions 6 – 8
- Allow about 53 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER:

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Question	1-5	6	7	8	Total
Inverse Functions	/4	/7	/7	/7	/25
Further Integration	/1	/5	/2	/5	/13
Further Trigonometry			/3		/3
	/5	/12	/12	/12	/41

Section I

5 marks

Attempt Questions 1-5

Allow about 7 minutes for this section

Use the multiple choice answer sheet for Questions 1-5.

1 What is the domain of the function $y = \arcsin\left(\frac{x}{5}\right)$?

A. $x \in \left[-\frac{1}{5}, \frac{1}{5}\right]$

B. $x \in [-5, 5]$

C. $x \in \left[-\frac{\pi}{10}, \frac{\pi}{10}\right]$

D. $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$

2 Which expression is an antiderivative of $\sin^2 3x$?

A. $-\frac{1}{3}\cos^2 3x$

B. $-\frac{\sin^3 3x}{\cos 3x}$

C. $\frac{x}{2} + \frac{1}{12}\sin 6x$

D. $\frac{x}{2} - \frac{1}{12}\sin 6x$

3 Which expression is equivalent to $\int \frac{dx}{5+4x^2}$?

A. $\frac{1}{2\sqrt{5}} \tan^{-1} \frac{2x}{\sqrt{5}} + c$

B. $\frac{1}{\sqrt{5}} \tan^{-1} \frac{2x}{\sqrt{5}} + c$

C. $\frac{2}{\sqrt{5}} \tan^{-1} \frac{2x}{\sqrt{5}} + c$

D. $\frac{1}{4} \tan^{-1} \frac{2x}{\sqrt{5}} + c$

4 Which of the following is a general solution for the equation $\tan 3x = \sqrt{3}$?

A. $x = \frac{\pi}{6} + \frac{2k\pi}{3}, x = \frac{7\pi}{6} + \frac{2k\pi}{3}, k \in \mathbb{Z}$

B. $x = \frac{\pi}{6} + 2k\pi, x = \frac{7\pi}{6} + 2k\pi, k \in \mathbb{Z}$

C. $x = \frac{\pi}{9} + \frac{2k\pi}{3}, x = \frac{4\pi}{9} + \frac{2k\pi}{3}, k \in \mathbb{Z}$

D. $x = \frac{\pi}{9} + 2k\pi, x = \frac{4\pi}{9} + 2k\pi, k \in \mathbb{Z}$

5. Consider the function $f(x) = \cos x + \sin x$, $\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$.

Which of the following is a correct equation for $f^{-1}(x)$?

A. $y = \sin^{-1} \frac{x}{\sqrt{2}} - \frac{\pi}{4}$

B. $y = \sin^{-1} \frac{x}{\sqrt{2}} + \frac{\pi}{4}$

C. $y = \cos^{-1} \frac{x}{\sqrt{2}} - \frac{\pi}{4}$

D. $y = \cos^{-1} \frac{x}{\sqrt{2}} + \frac{\pi}{4}$

Section II

36 marks

Attempt Questions 6 - 8

Allow about 53 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 6-8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (12 marks) Use a SEPARATE writing booklet

(a) Find the exact value of $\sin^{-1}\left(\sin \frac{4\pi}{3}\right)$. 1

(b) Differentiate $\sin^{-1}(x^2)$. 2

(c) Evaluate $\int_0^{\frac{4}{3}} \frac{dx}{\sqrt{16-9x^2}}$. 2

(d) Find $\int \cos 2x \sin 5x \, dx$. 2

(e) (i) Sketch the graph of $y = 12 \sin^{-1} \frac{x}{2}$, showing all important features. 2

(ii) The region in the first quadrant bounded by $y = 12 \sin^{-1} \frac{x}{2}$ and the line $y = 3\pi x$ is rotated around the y -axis to form a solid of revolution. 3

Find the volume of this solid.

End of Question 6

Question 7 (12 marks) Use a SEPARATE writing booklet

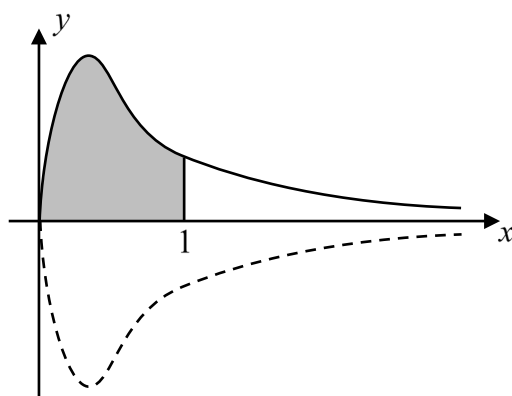
- (a) Use the substitution $u = \sqrt{x}$ to find $\int \frac{dx}{\sqrt{x}(x+1)}$. **2**
- (b) Find the exact value of $\sin\left(\sin^{-1}\left(\frac{3}{7}\right) + \cos^{-1}\left(\frac{1}{5}\right)\right)$. **3**
- (c) Given that $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$ find all solutions of the polynomial equation $8x^3 - 6x - 1 = 0$ in exact trigonometric form. **3**
- (d) Consider the function $f(x) = 2\sin^{-1}\sqrt{x} - \sin^{-1}(2x-1)$ for $0 \leq x \leq 1$.
- (i) Show that $f'(x) = 0$ for $0 < x < 1$. **2**
- (ii) Hence sketch the graph of $y = f(x)$. **2**

End of Question 7

Question 8 (12 marks) Use a SEPARATE writing booklet

(a) (i) Use the substitution $u = 3x + 1$ to show that $\int_0^1 \frac{x}{(3x+1)^2} dx = \frac{2}{9} \ln 2 - \frac{1}{12}$. 3

- (ii) The diagram shows the shaded region bounded by the curve $y = \frac{6\sqrt{x}}{3x+1}$, the x -axis, and the line $x = 1$. 2



This region is to be rotated about the x -axis to form a solid of revolution.

Use part (i) to find the volume of this solid in exact form.

(b) Use a trigonometric identity or otherwise to evaluate $\int_0^{\frac{1}{2}} \sin(2 \cos^{-1} x) dx$. 3

- (c) A runner and a cyclist compete in a race along a long straight running track. 4

They start at the same time at point S at the beginning of the track, each moving at their own constant speed, where the cyclist is three times as fast as the runner.

An observer stands at P , one metre from S , where SP is perpendicular to the track.

What is the greatest angle between the runner and the cyclist, as seen by the observer, as the race proceeds?

End of Paper

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2022 HSC Ext 1 Task 3 Solutions

(I) Multiple Choice

Summary: 1. **B** 2. **D** 3. **A** 4. **C** 5. **D**

1. $-1 \leq \frac{x}{5} \leq 1$
 $-5 \leq x \leq 5$ **(B)**

2. $\int \sin^2 3x \, dx = \frac{1}{2} \int (1 - \cos 6x) \, dx$
 $= \frac{1}{2} \left(x - \frac{\sin 6x}{6} \right)$
 $= \frac{x}{2} - \frac{\cos 6x}{12}$ **(D)**

3. $\int \frac{dx}{5+4x^2} = \frac{1}{4} \int \frac{dx}{\frac{5}{4} + x^2}$
 $= \frac{1}{4} \cdot \frac{2}{\sqrt{5}} \tan^{-1} \frac{x}{\sqrt{5}/2} + c$
 $= \frac{1}{2\sqrt{5}} \tan^{-1} \frac{2x}{\sqrt{5}} + c$ **(A)**

4. $\tan 3x = \sqrt{3}$
 $3x = \frac{\pi}{3} + 2k\pi, \frac{4\pi}{3} + 2k\pi, k \in \mathbb{Z}$
 $x = \frac{\pi}{9} + \frac{2k\pi}{3}, \frac{4\pi}{9} + \frac{2k\pi}{3}$ **(C)**

5. $y = \cos x + \sin x, 0 \leq x - \frac{\pi}{4} \leq \pi, \frac{\pi}{2} \leq x + \frac{\pi}{4} \leq \frac{3\pi}{2}$

$y = \sqrt{2} \cos(x - \frac{\pi}{4})$ OR $y = \sqrt{2} \sin(x + \frac{\pi}{4})$ [aux. angle method]

Inverse: $x = \sqrt{2} \cos(y - \frac{\pi}{4})$ OR $x = \sqrt{2} \sin(y + \frac{\pi}{4})$

$\frac{x}{\sqrt{2}} = \cos(y - \frac{\pi}{4})$ OR $\frac{x}{\sqrt{2}} = \sin(y + \frac{\pi}{4})$

$y - \frac{\pi}{4} = \cos^{-1} \frac{x}{\sqrt{2}}$

OR $y + \frac{\pi}{4} = \pi - \sin^{-1} \frac{x}{\sqrt{2}}$ [match domain]

$y = \cos^{-1} \frac{x}{\sqrt{2}} + \frac{\pi}{4}$

OR $y = -\sin^{-1} \frac{x}{\sqrt{2}} + \frac{3\pi}{4}$

(D)

Question 6

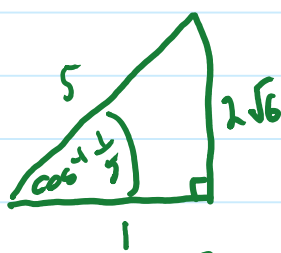
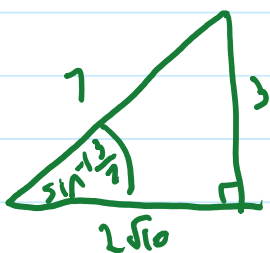
$$\begin{aligned} (a) \quad \sin \frac{4\pi}{3} &= -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2} \\ \sin^{-1}(\sin \frac{4\pi}{3}) &= \sin^{-1}(-\frac{\sqrt{3}}{2}) \\ &= -\sin^{-1}(\frac{\sqrt{3}}{2}) \\ &= \boxed{-\frac{\pi}{3}} \quad [1] \end{aligned}$$

$$\begin{aligned} (b) \quad \frac{d}{dx} \sin^{-1}(x^2) &= \frac{1}{\sqrt{1-x^2}} \times 2x \\ &= \boxed{\frac{2x}{\sqrt{1-x^2}}} \quad [2] \end{aligned}$$

$$\begin{aligned} (c) \quad \int_0^{4/3} \frac{dx}{\sqrt{16-9x^2}} &= \frac{1}{3} \int_0^{4/3} \frac{dx}{\sqrt{\frac{16}{9}-x^2}} \\ &= \frac{1}{3} \left[\sin^{-1} \frac{3x}{4} \right]_0^{4/3} \\ &= \frac{1}{3} (\sin^{-1} 1 - \sin^{-1} 0) \\ &= \boxed{\frac{\pi}{6}} \quad [2] \end{aligned}$$

OR $\int_0^{4/3} \frac{dx}{\sqrt{16-9x^2}} = \frac{1}{3} \int_0^{4/3} \frac{3x}{\sqrt{16-(3x)^2}} dx$
 $= \frac{1}{3} \left[\sin^{-1} \frac{3x}{4} \right]_0^{4/3}$

$$(d) \quad \sin(\sin^{-1} \frac{3}{7} + \cos^{-1} \frac{1}{5}) = \sin(\sin^{-1} \frac{3}{7}) \cos(\cos^{-1} \frac{1}{5}) + \cos(\sin^{-1} \frac{3}{7}) \sin(\cos^{-1} \frac{1}{5})$$



$$= \frac{3}{7} \times \frac{1}{5} + \frac{2\sqrt{10}}{7} \times \frac{2\sqrt{6}}{5}$$

$$= \frac{3}{35} + \frac{4\sqrt{60}}{35}$$

$$= \boxed{\frac{3+8\sqrt{15}}{35}} \quad [3]$$

$$(e) \cos A \sin B = \frac{1}{2} [\sin(A+B) - \sin(A-B)]$$

$$\begin{aligned} \cos 2x \sin 5x &= \frac{1}{2} [\sin(2x+5x) - \sin(2x-5x)] \\ &= \frac{1}{2} (\sin 7x + \sin 3x) \end{aligned}$$

$$\int \cos 2x \sin 5x \, dx = \frac{1}{2} \int (\sin 7x + \sin 3x) \, dx$$

$$= \boxed{-\frac{1}{14} \cos 7x - \frac{1}{6} \cos 3x + C} \quad [2]$$

$$\begin{aligned} (f) \int \frac{dx}{\sqrt{x}(x+1)} &= 2 \int \frac{1}{(\sqrt{x})^2 + 1} \cdot \frac{dx}{2\sqrt{x}} \\ &= 2 \int \frac{du}{u^2 + 1} \end{aligned}$$

$$\begin{aligned} u &= x^{1/2} \\ du &= \frac{1}{2} x^{-1/2} dx \\ &= \frac{1}{2\sqrt{x}} dx \end{aligned}$$

$$= 2 \tan^{-1} u + C$$

$$= \boxed{2 \tan^{-1}(\sqrt{x}) + C} \quad [3]$$

/13

Question 7

(a) (i) $f(x) = 2\sin^{-1}\sqrt{x} - \sin^{-1}(2x-1)$

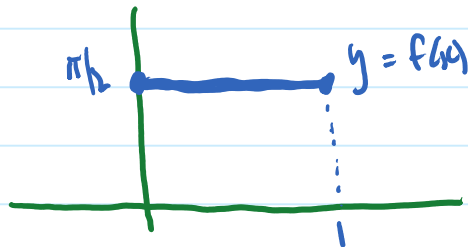
$$\begin{aligned}f'(x) &= \frac{2}{\sqrt{1-(\sqrt{x})^2}} \times \frac{1}{2\sqrt{x}} - \frac{1}{\sqrt{1-(2x-1)^2}} \times 2 \\&= \frac{1}{\sqrt{x} \cdot \sqrt{1-x}} - \frac{2}{\sqrt{1-(4x^2-4x+1)}} \\&= \frac{1}{\sqrt{x-x^2}} - \frac{2}{\sqrt{4x-4x^2}} \\&= \frac{1}{\sqrt{x-x^2}} - \frac{1}{\sqrt{x-x^2}} \\&= 0\end{aligned}$$

[2]

(ii) There are no discontinuities in $[0, 1]$.

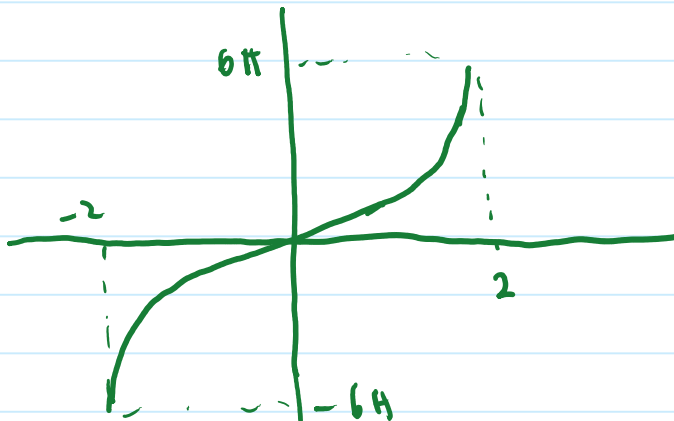
$$\begin{aligned}f(0) &= 2\sin^{-1}0 - \sin^{-1}(-1) \\&= \frac{\pi}{2}\end{aligned}$$

As $f'(x) = 0$, and no discontinuities, $f(x)$ is a constant function



[2]

(b) (i)



[2]

$$(11) \quad y = 3\pi x \Rightarrow x = 2, y = 6\pi$$

$$y = 12 \sin^{-1} \frac{x}{2} \Rightarrow \frac{y}{12} = \sin^{-1} \frac{x}{2}$$

$$\frac{x}{2} = \sin \frac{y}{12}$$

$$x = 2 \sin \frac{y}{12}$$

$$V = \pi \int_0^{6\pi} \left(2 \sin \frac{y}{12}\right)^2 dy - \frac{1}{3} \pi \times 2^2 \times 6\pi \quad (\text{Vol. of cone: } \frac{1}{3} \pi r^2 h)$$

$$= 4\pi \int_0^{6\pi} \sin^2 \frac{y}{12} dy - 8\pi^2$$

$$= 2\pi \int_0^{6\pi} (1 - \cos \frac{y}{6}) dy - 8\pi^2$$

$$= 2\pi \left[y - 6 \sin \frac{y}{6} \right]_0^{6\pi} - 8\pi^2$$

$$= 2\pi [6\pi - 0] - 8\pi^2$$

$$= \boxed{4\pi^2 \text{ units}^3} \quad [3]$$

$$(c) \quad \text{let } x = \sin \theta \quad 8x^3 - 6x - 1 = 0 \Rightarrow 8\sin^3 \theta - 6\sin \theta - 1 = 0$$

$$-2(3\sin \theta - 4\sin^3 \theta) = 1$$

$$\sin 3\theta = -\frac{1}{2}$$

$$3\theta = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{6}, \frac{23\pi}{6}, \frac{31\pi}{6}, \frac{35\pi}{6}, \quad 3\theta \in [0, 6\pi]$$

$$\theta = \frac{7\pi}{18}, \frac{11\pi}{18}, \frac{19\pi}{18}, \frac{23\pi}{18}, \frac{31\pi}{18}, \frac{35\pi}{18}, \quad \theta \in [0, 2\pi]$$

$$x = \sin \frac{7\pi}{18}, \sin \frac{11\pi}{18}, \sin \frac{19\pi}{18}, \sin \frac{23\pi}{18}, \sin \frac{31\pi}{18}, \sin \frac{35\pi}{18}$$

$$\text{but } \sin \frac{7\pi}{18} = \sin \frac{11\pi}{18}, \sin \frac{19\pi}{18} = \sin \frac{35\pi}{18}, \sin \frac{23\pi}{18} = \sin \frac{31\pi}{18}$$

$$\therefore \boxed{x = \sin \frac{7\pi}{18}, \sin \frac{19\pi}{18}, \sin \frac{23\pi}{18}} \quad [3]$$

$$\text{alternatively, } x = \sin \frac{7\pi}{18}, -\sin \frac{\pi}{18}, -\sin \frac{5\pi}{18}$$

$$\text{OR } \sin 70^\circ, \sin 190^\circ, \sin 230^\circ$$

$$\text{OR } \sin 70^\circ, -\sin 10^\circ, -\sin 50^\circ$$

12

Question 8

$$(a) \quad (i) \int_0^1 \frac{x \, dx}{(3x+1)^2} = \int_1^4 \frac{1}{u^2} \cdot \frac{1}{3}(u-1) \cdot \frac{1}{3} \, du$$

$$= \frac{1}{9} \int_1^4 \frac{u-1}{u^2} \, du$$

$$= \frac{1}{9} \int_1^4 \left(\frac{1}{u} - \frac{1}{u^2} \right) \, du$$

$$= \frac{1}{9} \left[\ln u + \frac{1}{u} \right]_1^4$$

$$= \frac{1}{9} \left\{ \left(\ln 4 + \frac{1}{4} \right) - (0+1) \right\}$$

$$= \frac{1}{9} \left(2\ln 2 - \frac{3}{4} \right)$$

$$= \frac{2}{9} \ln 2 - \frac{1}{12} \quad [3]$$

$$\text{let } u = 3x+1$$

$$du = 3 \, dx$$

$$x=0, \quad u=1$$

$$x=1, \quad u=4$$

$$x = \frac{1}{3}(u-1)$$

$$(ii) \quad V = \pi \int_0^1 \left(\frac{6\sqrt{x}}{3x+1} \right)^2 \, dx$$

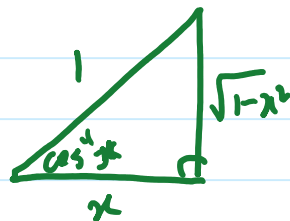
$$= 36\pi \int_0^1 \frac{x}{(3x+1)^2} \, dx$$

$$= 36\pi \times \left(\frac{2}{9} \ln 2 - \frac{1}{12} \right)$$

$$= \boxed{\pi(8\ln 2 - 3) \text{ units}^3} \quad [2]$$

$$(b) \quad \sin(2\cos^{-1}x) = 2\sin(\cos^{-1}x) \cos(\cos^{-1}x)$$

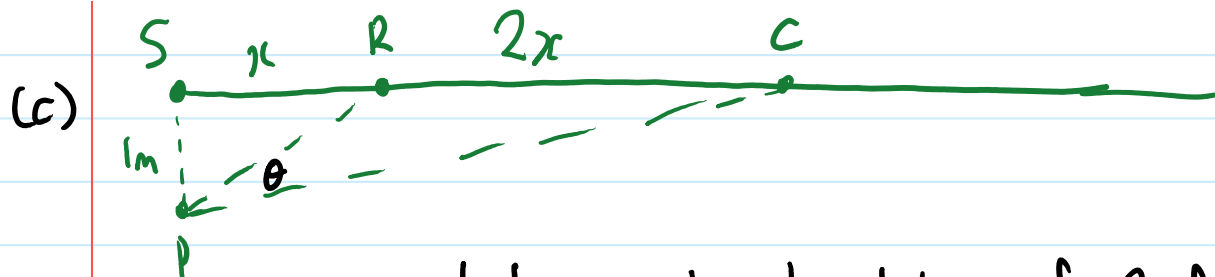
$$= 2 \cdot \sqrt{1-x^2} \cdot x$$



$$\int_0^{1/\sqrt{2}} \sin(2\cos^{-1}x) \, dx = - \int_0^{1/\sqrt{2}} (1-x^2)^{1/2} \cdot (-2x) \, dx$$

$$= -\frac{2}{3} \left[(1-x^2)^{3/2} \right]_0^{1/\sqrt{2}}$$

$$= -\frac{2}{3} \left(\frac{3\sqrt{3}}{8} - 1 \right) = \boxed{\frac{1}{12} (8 - 3\sqrt{3})} \quad [3]$$



Let x metres be distance of R from S at same time after starting
 $\therefore 3x$ metres is distance of C from S at that same time.
 Let θ be angle between R and C as seen from P
 let $\alpha = \widehat{SPC}$, $\beta = \widehat{SPR}$

$$\tan \alpha = \frac{3x}{1} = 3x, \quad \tan \beta = x$$

$$\alpha = \tan^{-1} 3x \quad \beta = \tan^{-1} x$$

$$\begin{aligned} \theta &= \alpha - \beta \\ &= \tan^{-1} 3x - \tan^{-1} x \end{aligned}$$

$$\begin{aligned} \frac{d\theta}{dx} &= \frac{3}{1+9x^2} - \frac{1}{1+x^2} \\ &= \frac{3(1+x^2) - (1+9x^2)}{(1+9x^2)(1+x^2)} \\ &= \frac{2-6x^2}{(1+9x^2)(1+x^2)} \end{aligned}$$

For max/min, $\frac{d\theta}{dx} = 0 \Rightarrow 6x^2 = 2$
 $x = \frac{1}{\sqrt{3}} \quad (x > 0)$

As $(1+9x^2)(1+x^2) > 0 \quad \forall x$, sign of $\frac{d\theta}{dx}$ is sign of $2-6x^2$

if $x < \frac{1}{\sqrt{3}}$, $x^2 < \frac{1}{3}$, $6x^2 < 2 \Rightarrow 2-6x^2 > 0 \Rightarrow \frac{d\theta}{dx} > 0$

if $x > \frac{1}{\sqrt{3}}$, $x^2 > \frac{1}{3}$, $6x^2 > 2 \Rightarrow 2-6x^2 < 0 \Rightarrow \frac{d\theta}{dx} < 0$

$\therefore \max \theta$ when $x = \frac{1}{\sqrt{3}}$

$$\begin{aligned} \theta_{\max} &= \tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \\ &= \frac{\pi}{3} - \frac{\pi}{6} \\ &= \boxed{\frac{\pi}{6}} \end{aligned}$$

(4)

12